Statistical analysis of diabetes and lifestyle indicators

BMAT202P Probability and Statistics Lab
CDC BRFSS 2015 with experimental design supplements

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Aim and course coverage

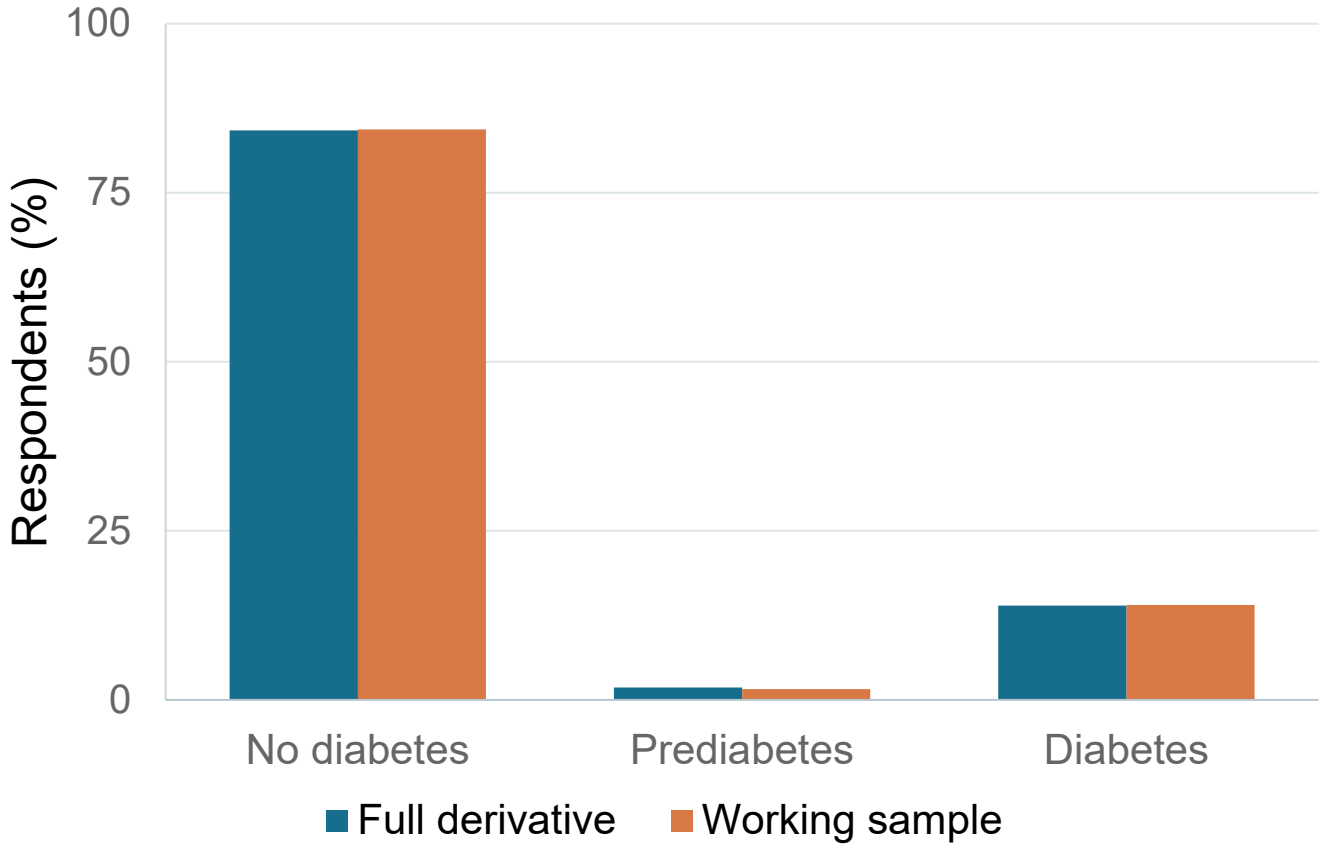
Aim: describe BMI and health indicators, examine their associations, and apply the statistical methods prescribed in BMAT202P.

Questions: How does BMI vary across health groups? How much BMI variation do the regression models explain? Which probability models fit the observations?

The course also requires independent and paired tests, chi-square tests, CRD, RBD and Latin Square ANOVA.

The health study is observational. The design supplements use separate experimental data.

Dataset source and working sample



The derivative contains 253,680 respondents, 21 predictors and one three-class target.
We sampled 3,000 records without replacement using seed 101.
Combined positive status is 15.63% in the sample and 15.76% in the full derivative.

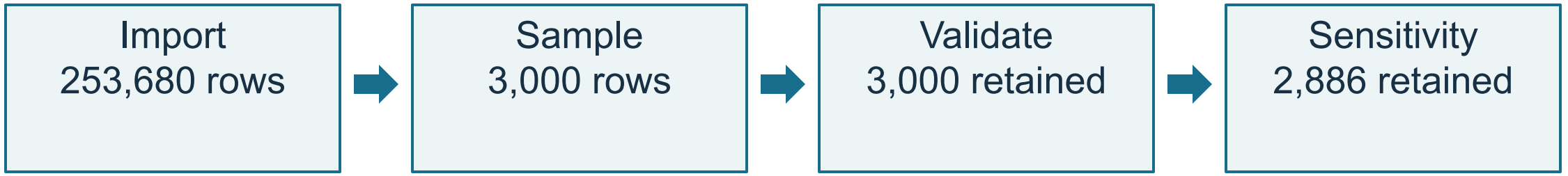
These are unweighted dataset proportions. They do not estimate national prevalence.

Variable definitions

The codebook distinguishes measurements, counts and coded categories.

Variable	Meaning	Statistical type
Diabetes_012	No diabetes / prediabetes / diabetes	Categorical outcome
Diabetes_binary	Prediabetes OR diabetes = Yes	Binary outcome
BMI	Body mass index in kg/m²	Continuous measure
MentHlth / PhysHlth	Poor-health days in a 30-day window	Bounded count
Age / Income / Education	Bands with unequal spacing	Ordinal categories
Smoker	Ever smoked at least 100 cigarettes	Binary indicator

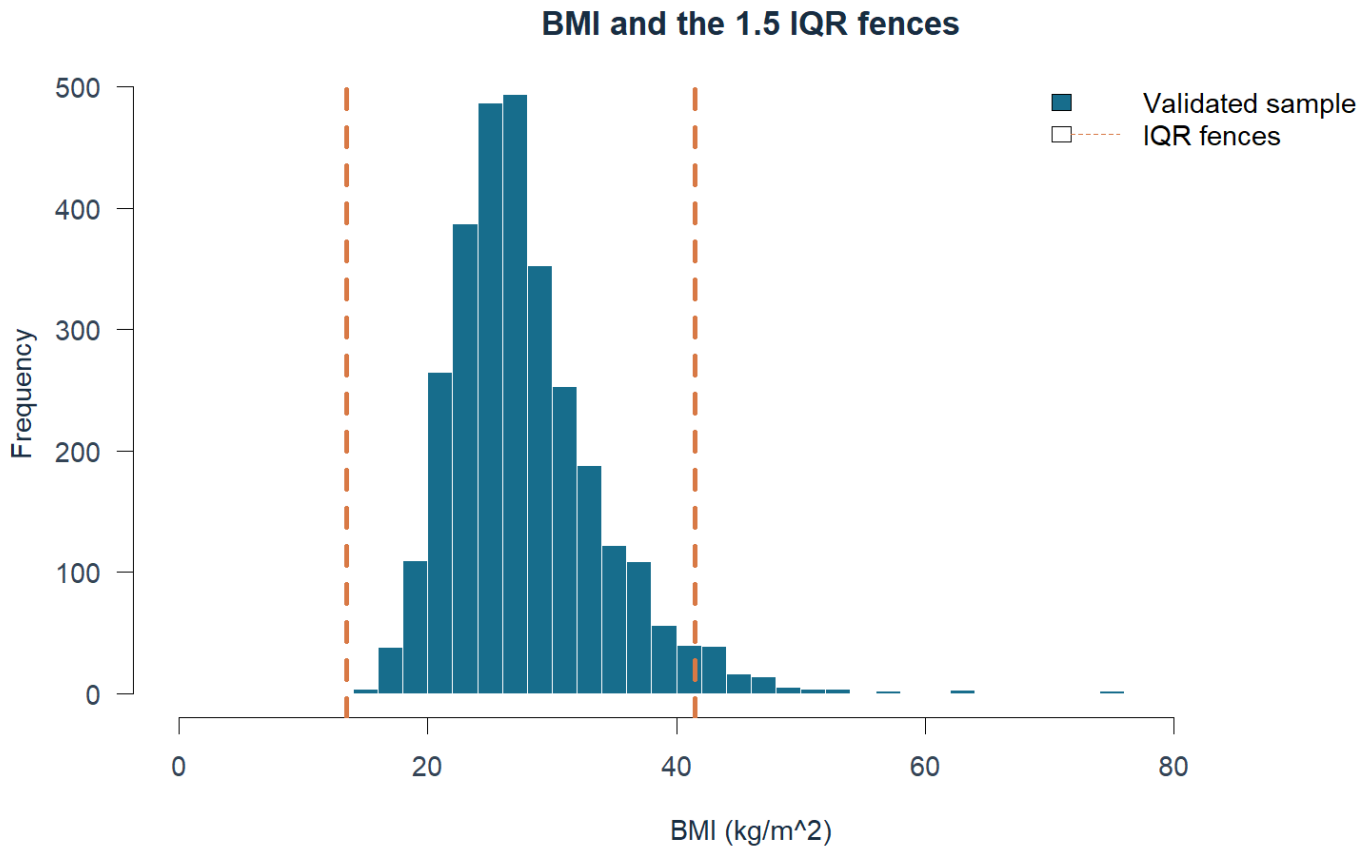
Data preparation



Validation checks column names, allowed codes, missing values and measurement ranges.
No invalid or missing records occurred in the 3,000-person working sample.
We preserve source-row identifiers and factor types in RDS. Identical feature profiles can belong to different people, so we retain them.

The original sample remains available alongside the separate BMI-filtered dataset.

BMI outlier analysis

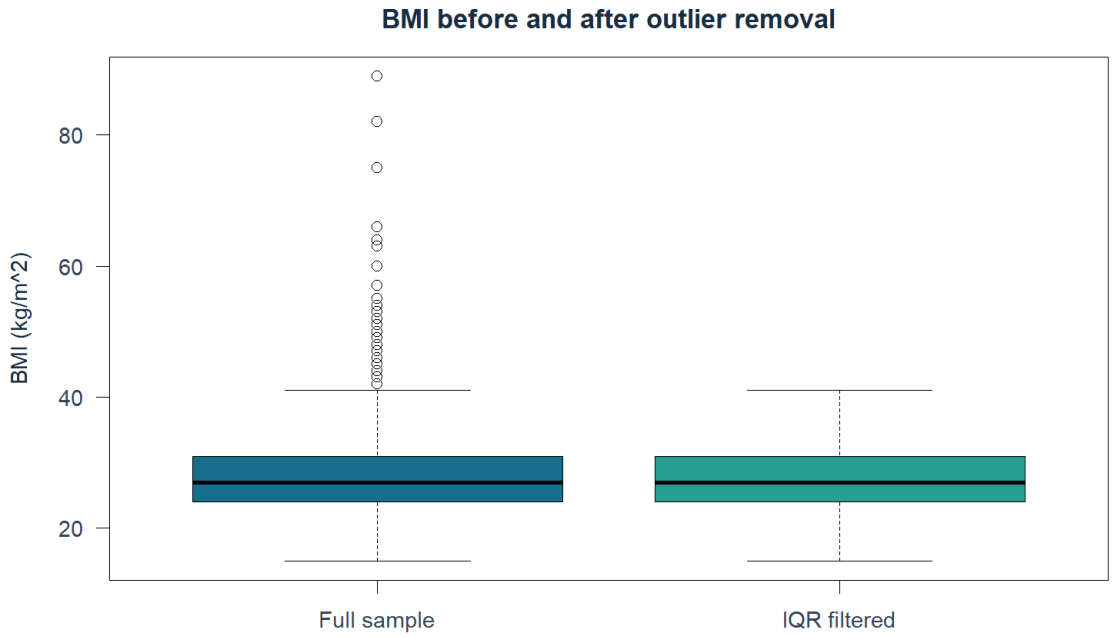


Q1 = 24.0 and Q3 = 31.0, giving IQR = 7.0.
The 1.5 IQR fences are 13.5 and 41.5.
114 respondents (3.8%) fall outside the fences.

An unusual but valid BMI is not automatically a recording error.

Outlier removal and its effect

We removed the 114 BMI flags only in a sensitivity dataset. The primary analysis retains all valid records.

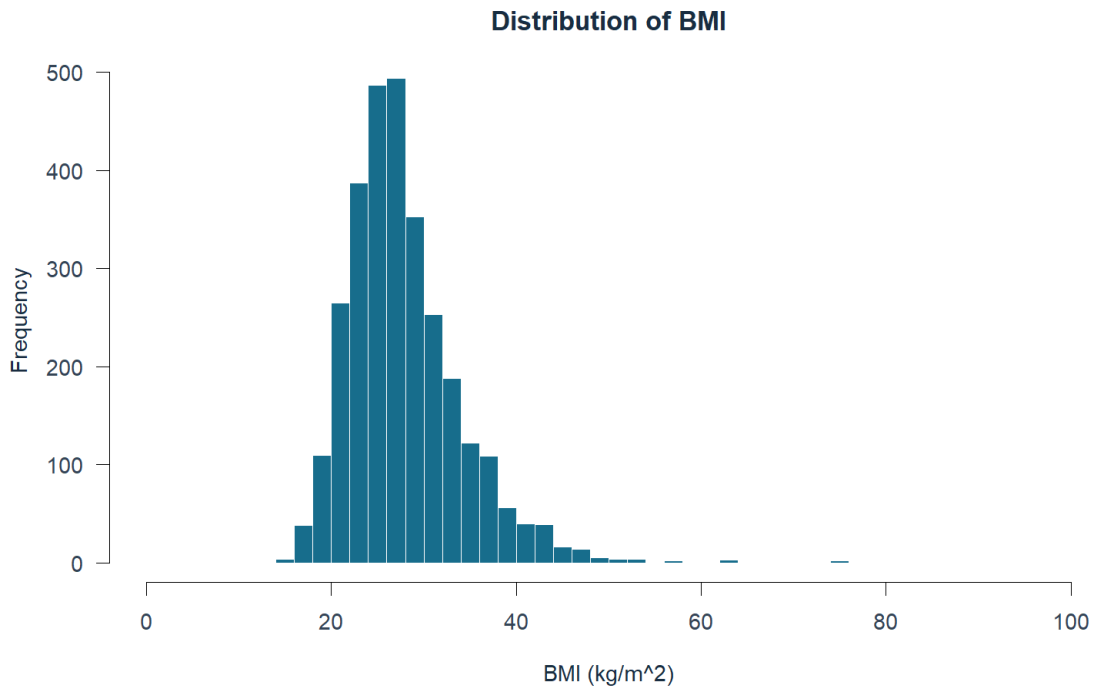


Measure	Full sample	Filtered
Respondents	3,000	2,886
Mean BMI	28.40	27.65
BMI variance	40.34	24.59
Positive status	15.63%	14.62%
Pearson r	0.102	0.056

Filtering lowers both average BMI and the positive-status proportion. It changes the sample composition.

Summary statistics and variance

BMI has a right tail. Poor-health day counts have median zero and substantial variation.

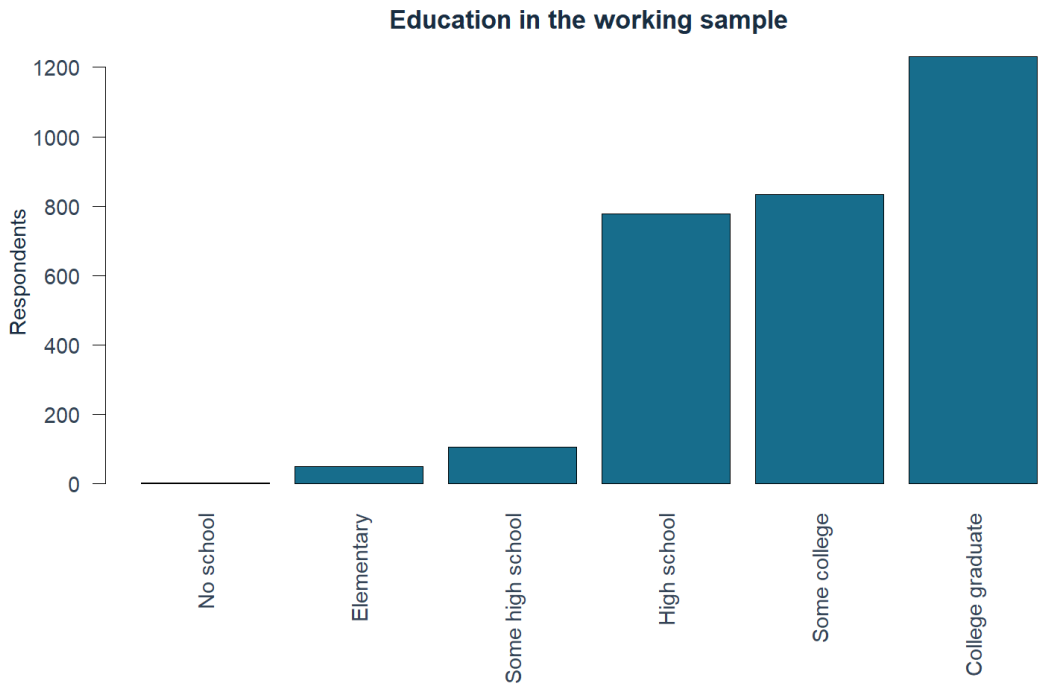


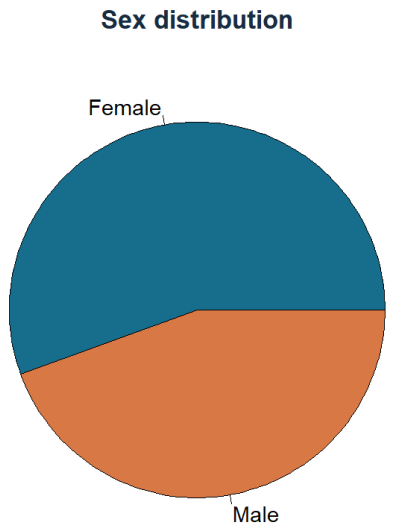
Variable	Mean	Median	SD	Var.
BMI	28.40	27	6.35	40.34
Mental days	3.26	0	7.39	54.67
Physical days	4.15	0	8.62	74.24

Variance measures squared spread. Standard deviation returns that spread to the original units.

Graphical representations of categories

Bar charts compare education counts. A pie chart demonstrates the course requirement for categorical proportions.

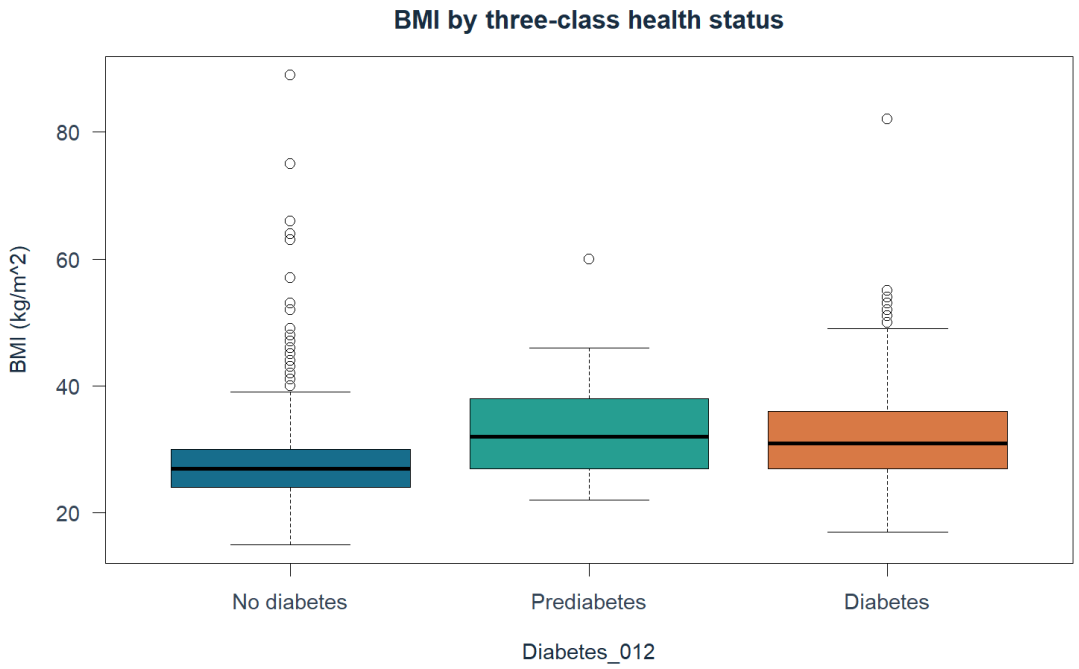


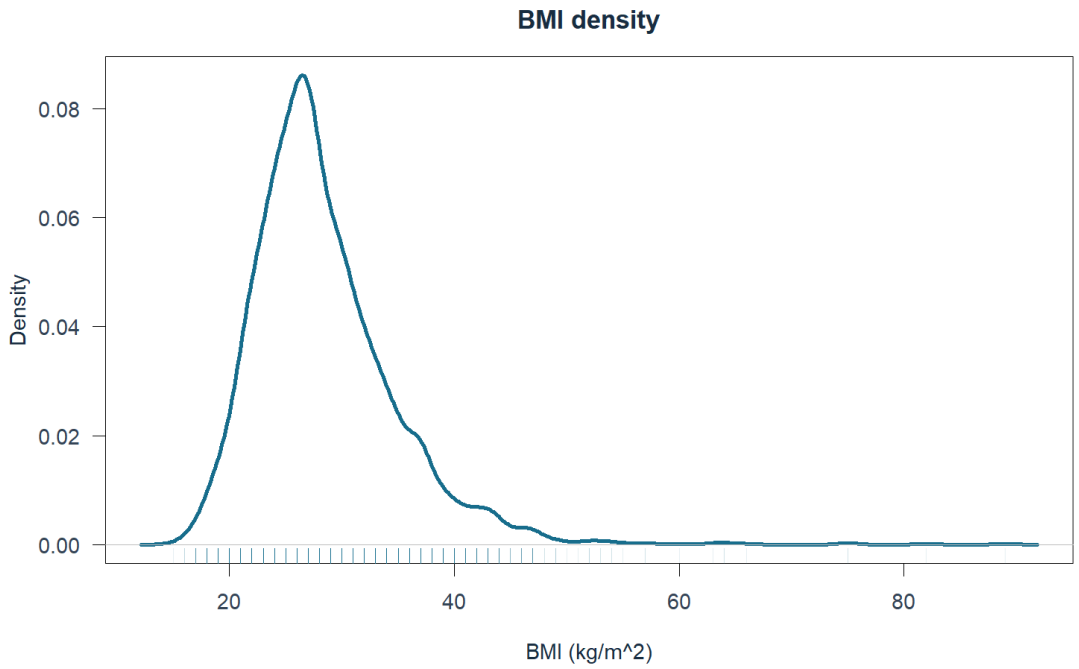


Group sizes matter. The no-school education category contains only two respondents.

BMI across health-status groups

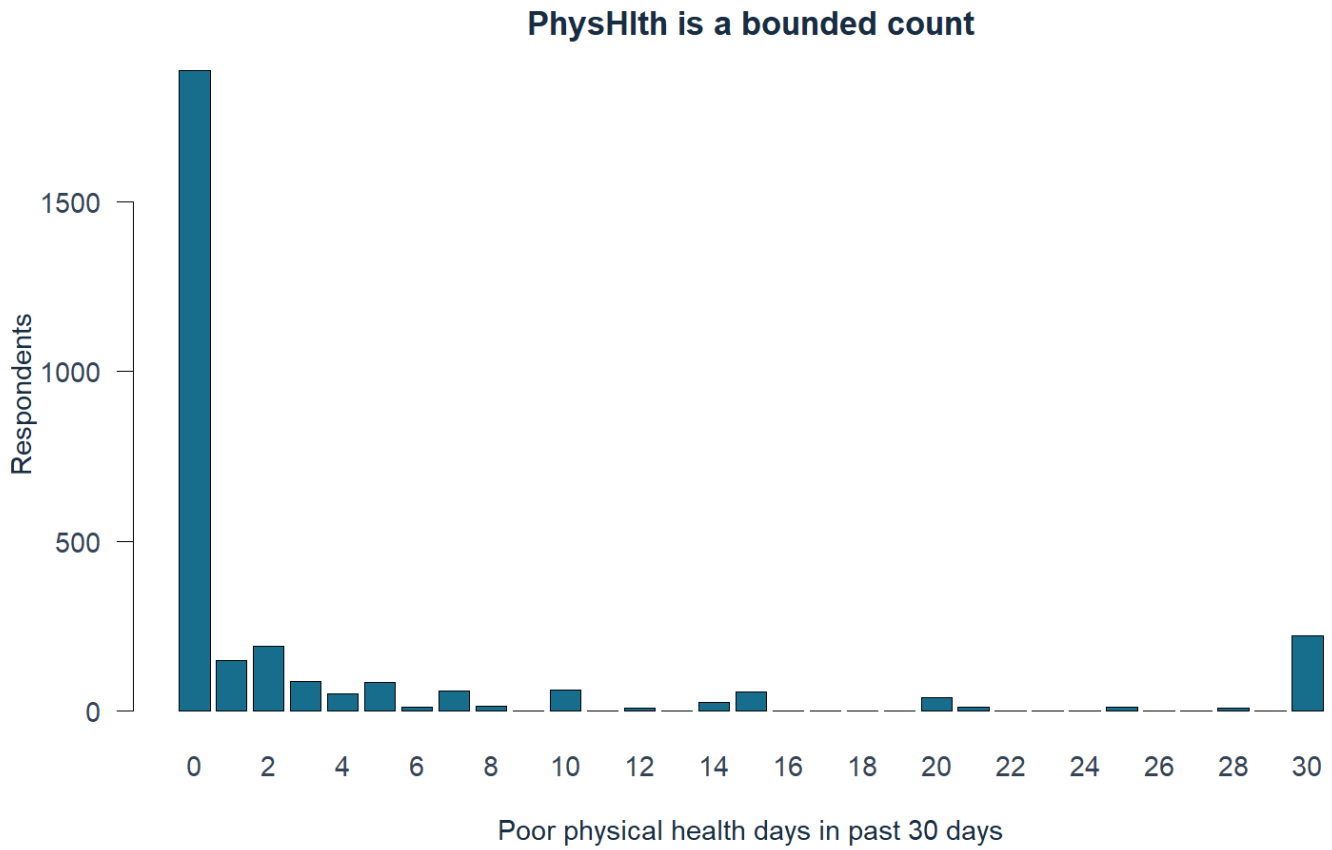
Mean BMI is 27.72 for no diabetes, 32.85 for prediabetes and 31.98 for diabetes. The distributions overlap.





The prediabetes group has only 48 respondents. Group differences describe associations.

Poor physical health days

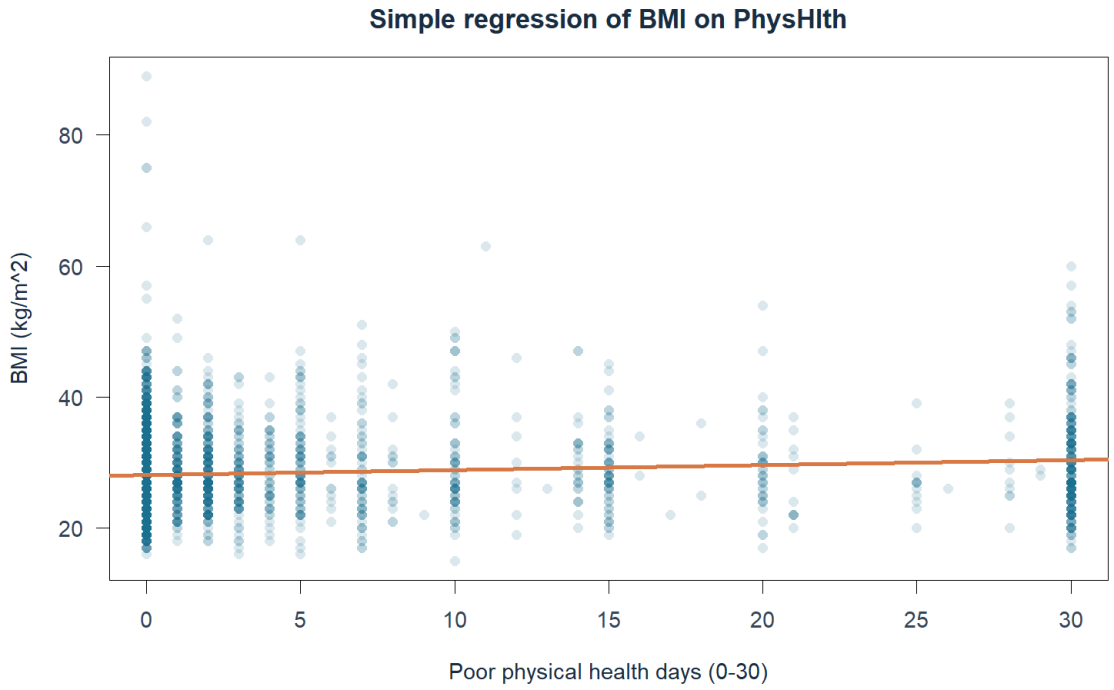


62.9% report zero poor-health days.
7.4% report all 30 days.
The mean is 4.15 days, while variance is 74.24 days².

The spikes at zero and 30 are important features of this bounded count.

Correlation and simple regression

Pearson r = 0.102, with 95% CI 0.066 to 0.137.
BMI = 28.09 + 0.0749 × PhysHlth.



Statistic	Result
Pearson r	0.102
Spearman r	0.108
Slope CI	0.0486 to 0.1012
R^2	1.03%
Slope p	<0.001

A significant association explains only 1.03% of BMI variation.

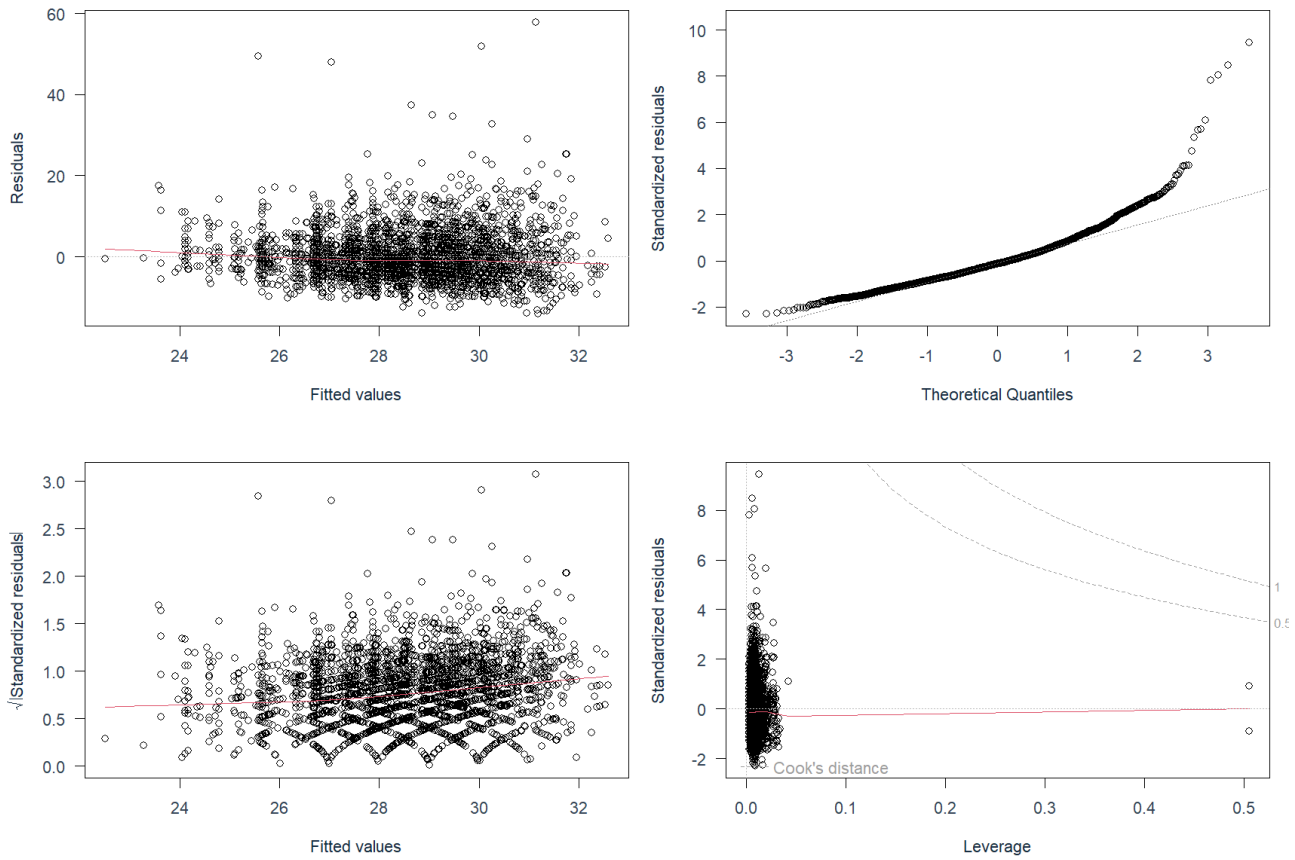
Multiple linear regression

Response: BMI. Predictors: age band, physical activity, income band, education and ever-smoking.
R^2 = 6.81%. Adjusted R^2 = 5.99%.
Joint tests compare each categorical predictor while keeping the others in the model.

Predictor	df	Partial F	p
Age	12	9.17	<0.001
PhysActivity	1	13.30	<0.001
Income	7	4.43	<0.001
Education	5	8.03	<0.001
Smoker	1	0.07	0.786

The model leaves most BMI variation unexplained. Statistical adjustment does not establish causation.

Regression diagnostics and sensitivity

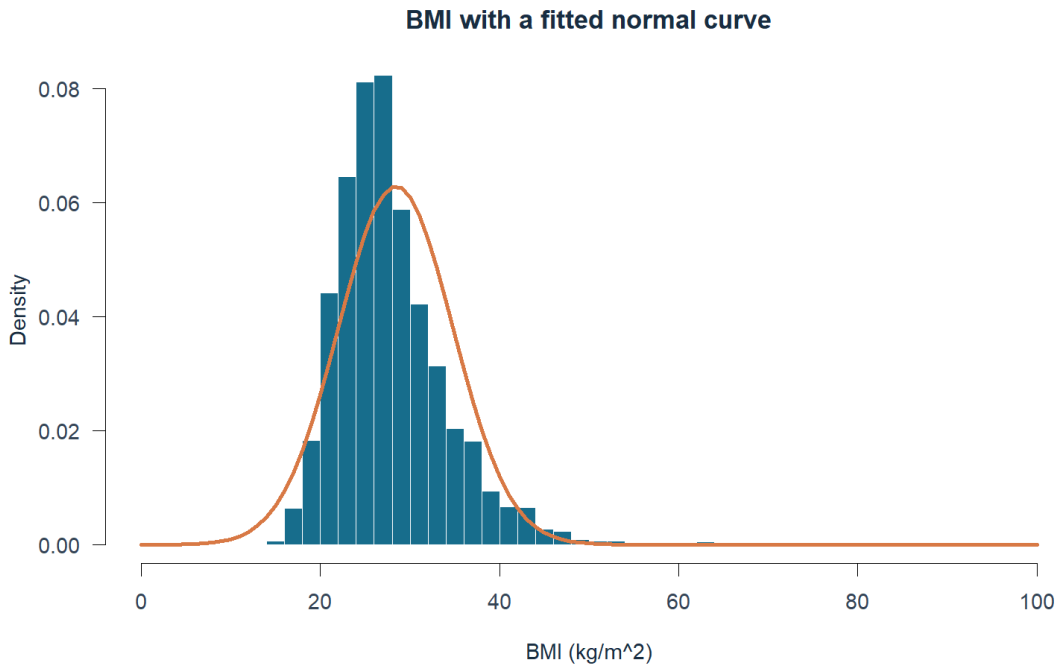


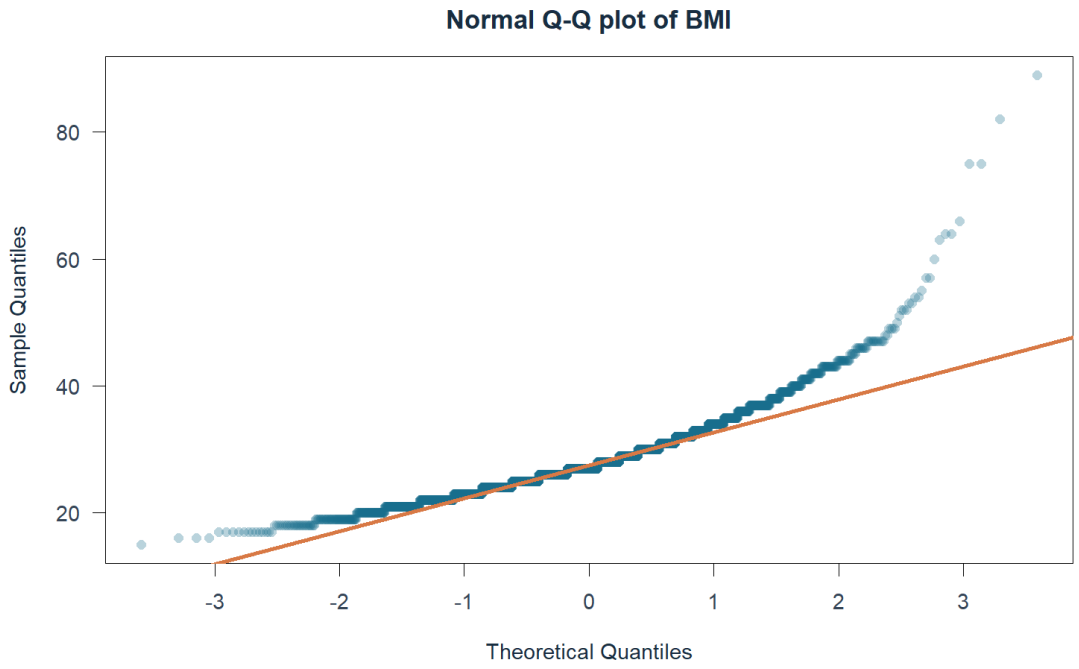
Residual plots assess linear form and changing spread. The Q-Q plot checks residual shape.
Leverage and Cook's distance identify observations that may influence the fitted coefficients.
After BMI filtering, adjusted R^2 changes from 5.99% to 4.89%.

Valid extreme observations affect the fit. Model conclusions remain cautious and descriptive.

Fitting the normal distribution

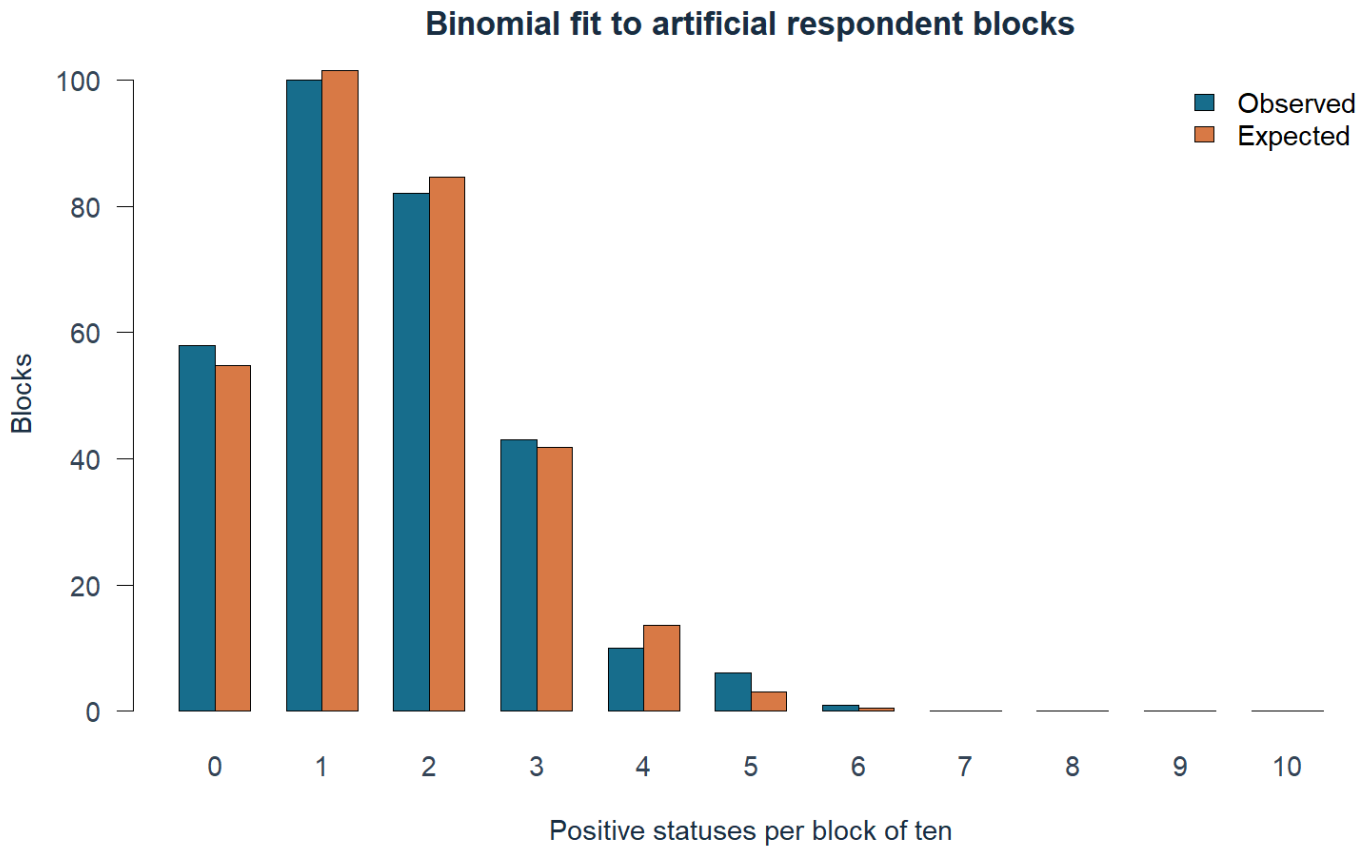
The fitted normal model uses mean 28.40 and SD 6.35.
Shapiro-Wilk gives W = 0.897 and p < 0.001 for all 3,000 BMI values.





The right tail departs from a normal distribution. Model residuals need their own checks.

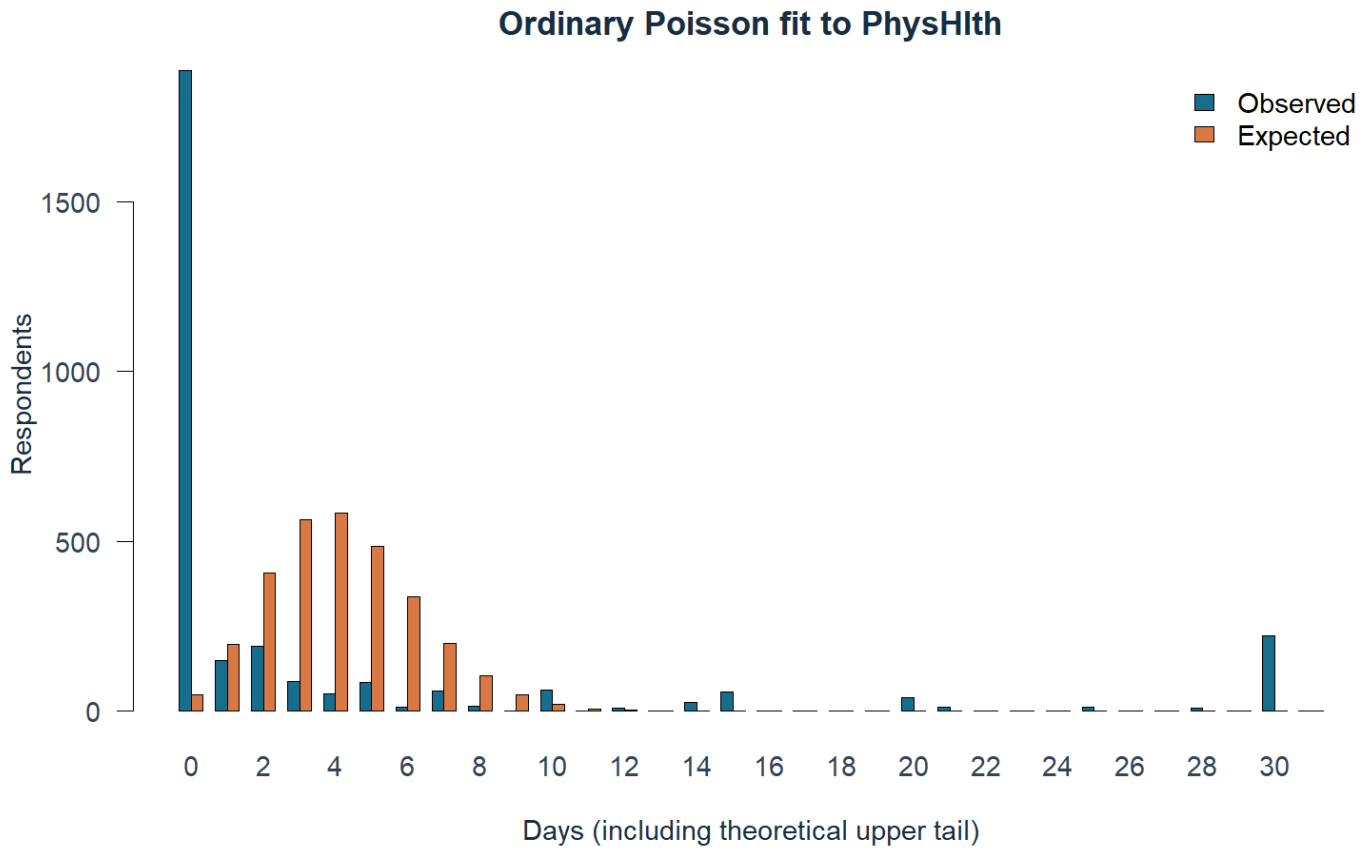
Fitting the binomial distribution



We count combined-positive statuses in 300 artificial blocks of ten.
The individual sample proportion is 0.1563.
After pooling and fitting to bin counts, $\chi^2 = 0.287$, df = 3, $p = 0.962$.

The pooled test does not reject the binomial teaching model. It does not prove identical individual risk.

Fitting the Poisson distribution



The ordinary Poisson mean estimate is $\lambda = 4.15$.
Variance/mean = 17.88, far above the Poisson value of one.
The pooled test refits $\lambda = 1.80$ and gives $\chi^2 = 41781.7$, $df = 6$, $p < 0.001$.

The Poisson model poorly describes these bounded, zero-heavy day counts.

One-sample hypothesis tests

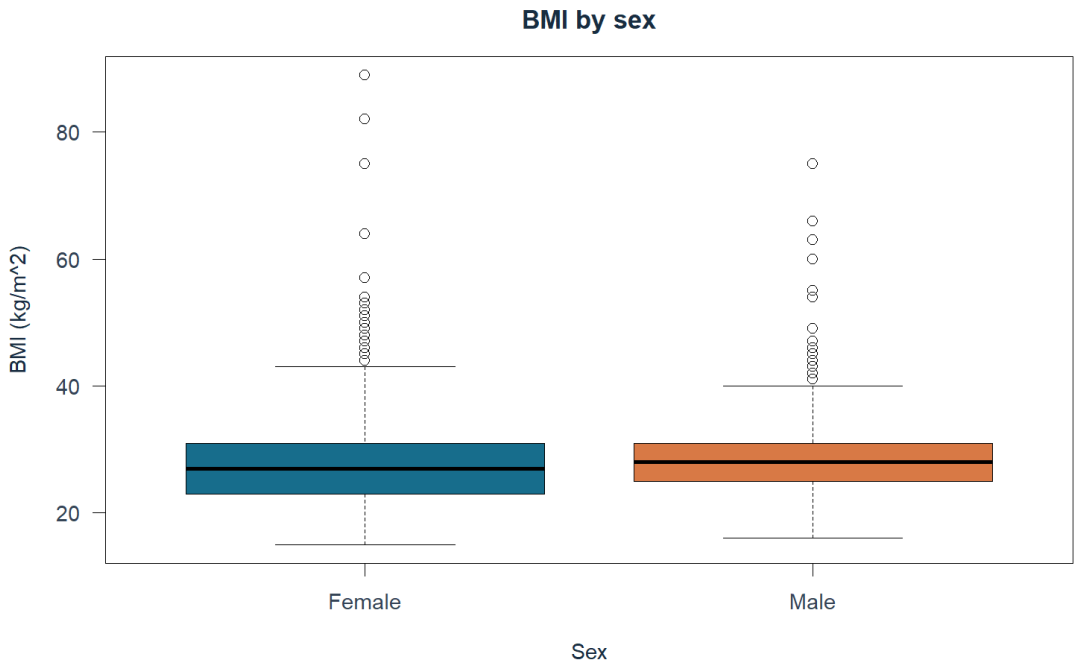
We use two-sided alternatives and $\alpha = 0.05$. The values 27 and 11% are hypothetical classroom benchmarks.

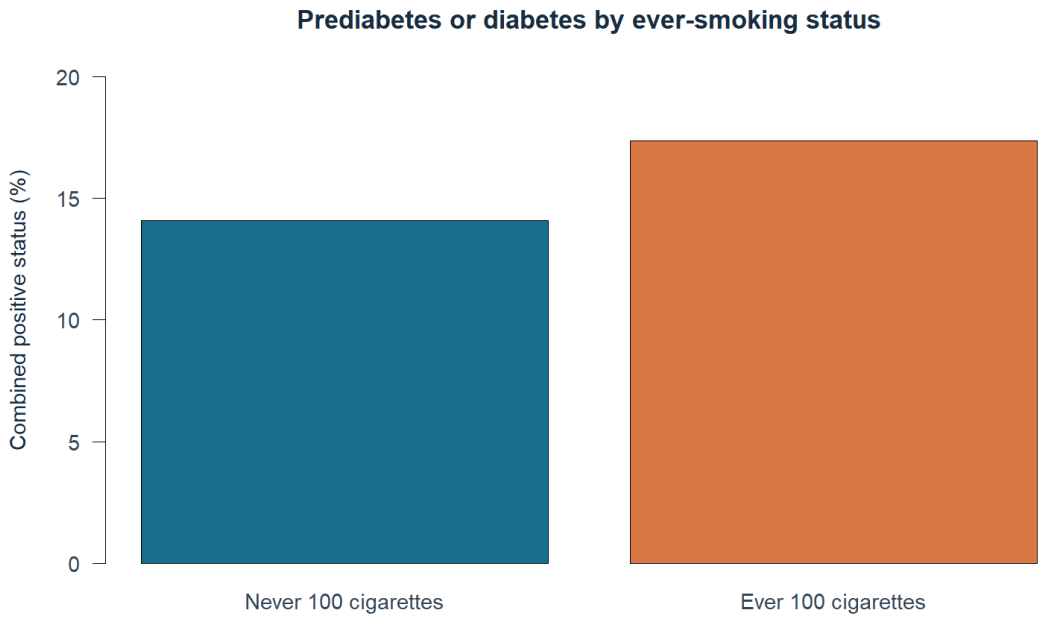
Question	Estimate and 95% CI	Statistic	p
Mean BMI = 27	28.40 [28.17, 28.63]	t = 12.06	<0.001
Positive proportion = 11%	15.63% [14.36%, 16.99%]	$\chi^2 = 65.31$	<0.001

Both teaching nulls are rejected. The 11% value is not a sourced national prevalence claim.

Two-sample and independent comparisons

Female minus male mean BMI is −0.65, with 95% CI −1.10 to −0.20 (p = 0.004).
Combined-positive proportions are 14.09% for never-smoking and 17.36% for ever-smoking (p = 0.016).

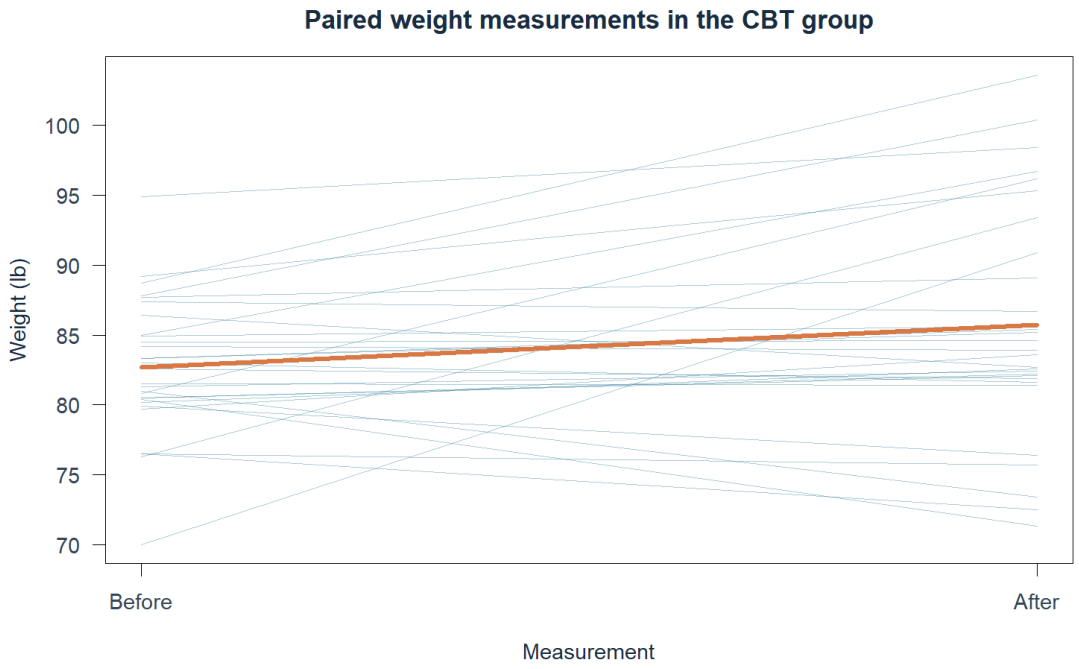




The independent PhysHlth test finds 1.25 more poor-health days for females, CI 0.63 to 1.86.

Paired t-test supplement

The CBT group in MASS::anorexia has 29 paired before/after weights.
Mean change is +3.01 lb, with 95% CI +0.23 to +5.79 lb. The paired t-test gives p = 0.035.
The signed-rank sensitivity test gives p = 0.064; the evidence depends on the method.

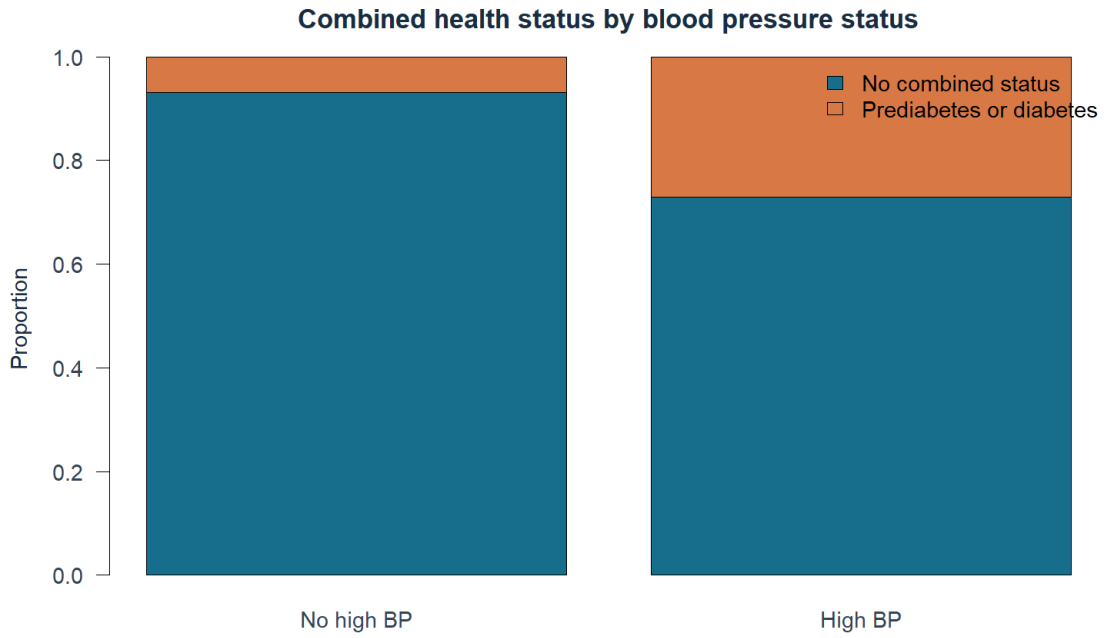




Pairing compares each patient with herself. The difference distribution is non-normal, so interpretation is cautious.

Chi-square tests and categorical association

Combined-positive status is 6.93% among respondents without high BP and 27.06% among those with high BP.

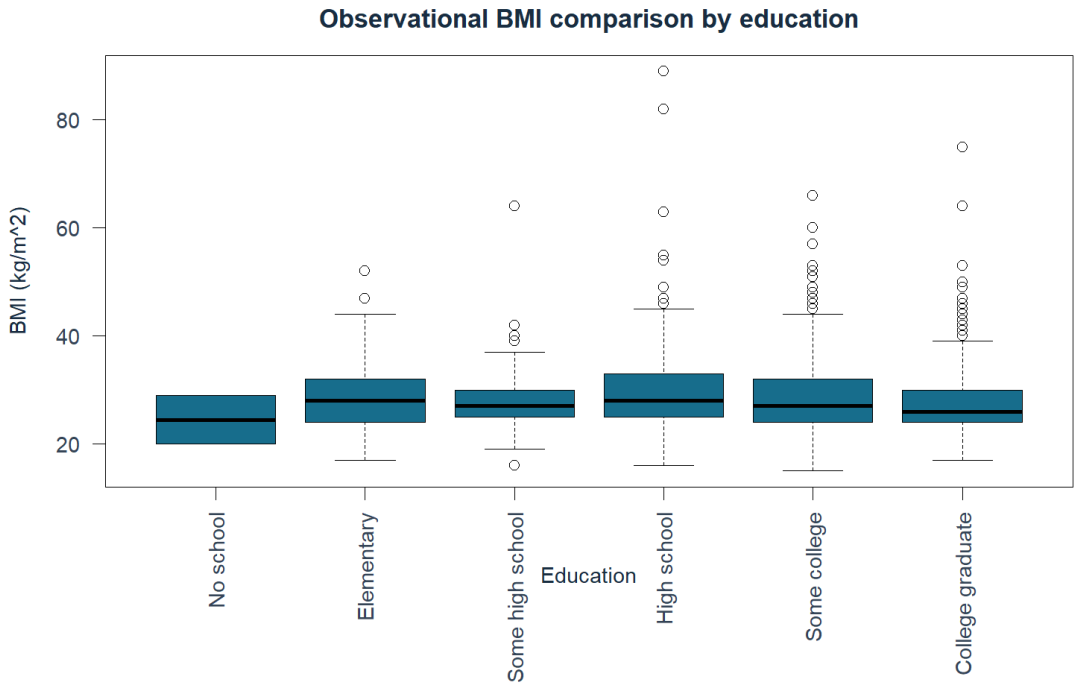


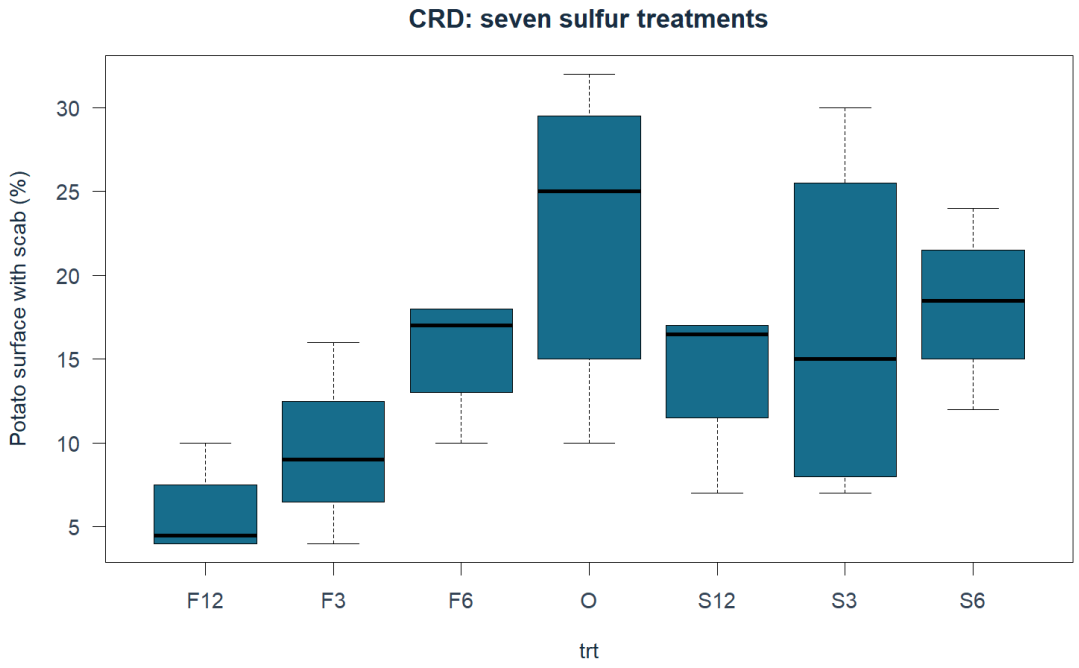
Test	χ^2	Effect size	p
Status and HighBP	226.28	V = 0.275	<0.001
Status and Smoker	6.07	V = 0.045	0.014
Uniform age-band GOF	678.12	df = 12	<0.001

HighBP has the stronger sample association. The age test uses a hypothetical equal-probability null.

One-way ANOVA and CRD

Education-group BMI differs in the sample (F = 14.41). Unequal variances motivate the Welch sensitivity result (p < 0.001).
The separate sulfur experiment is a genuine CRD: 32 plots and seven treatments (F = 3.61, p = 0.010).

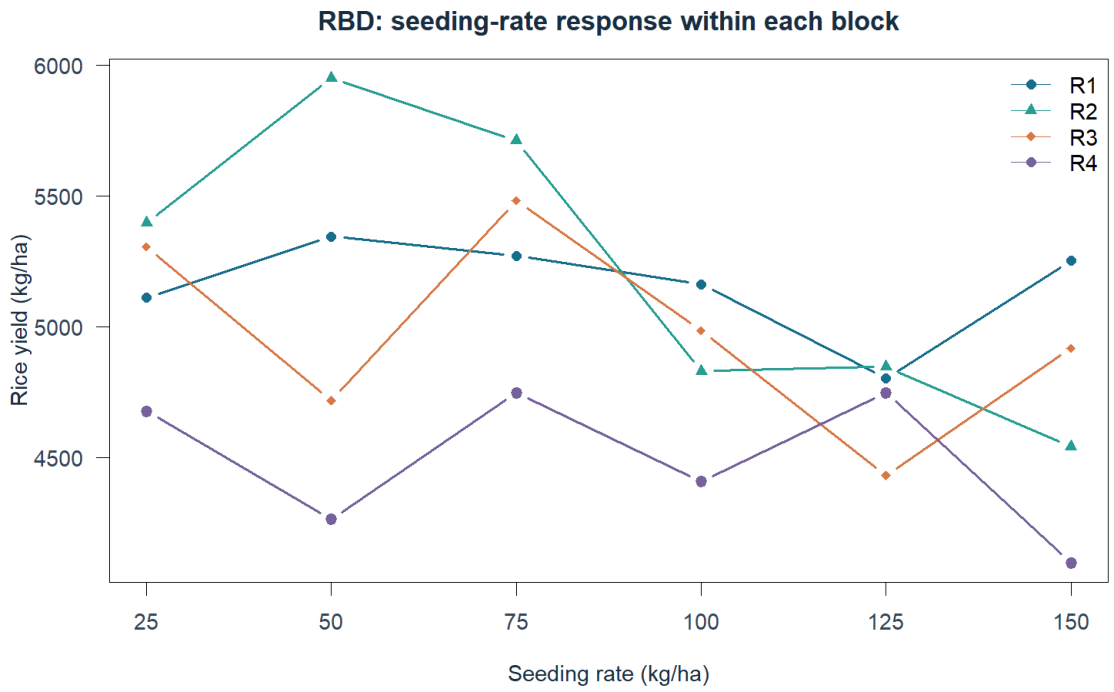




The education comparison is observational. The sulfur dataset supports the randomized-design exercise.

Randomized block design

The rice experiment includes six seeding rates in each of four complete blocks. The additive model separates block and rate effects.

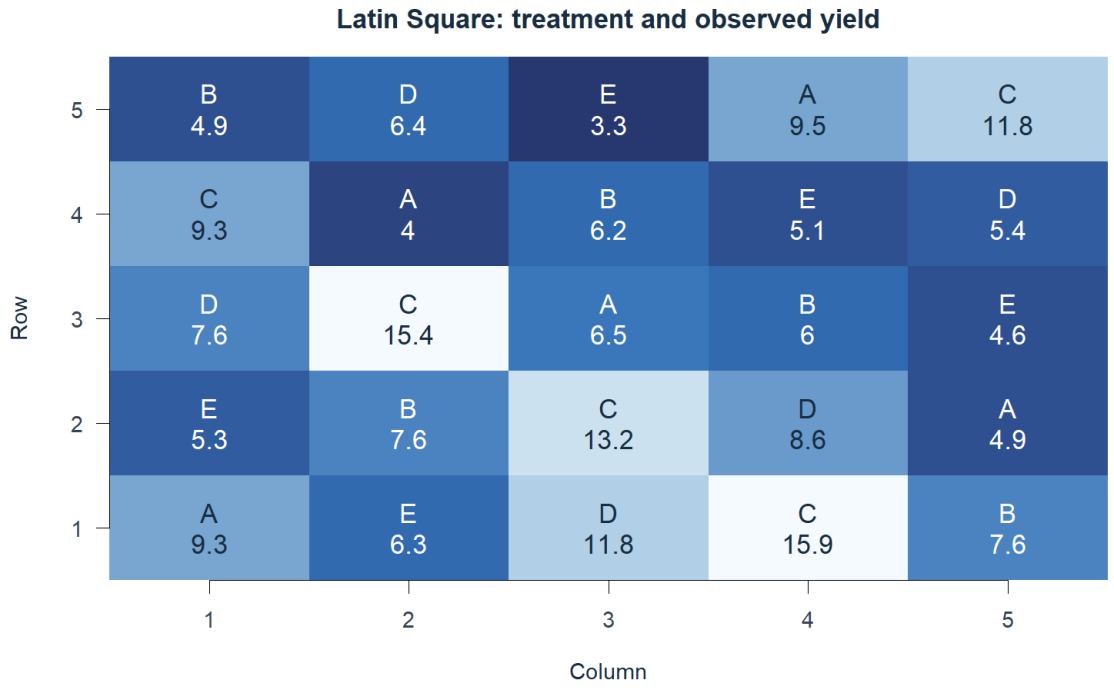


Term	df	F	p
Block	3	5.86	0.007
Seeding rate	5	2.17	0.113
Residual	15	MSE = 110,558	

Blocking lowers residual MSE by 44.8%. The rate effect is not significant at α = 0.05.

Latin Square design

Five wheat treatments occur once in every row and every column. The model controls two nuisance factors.



Term	df	p
Row	4	0.013
Column	4	0.263
Treatment	4	<0.001
Residual	12	

Weekly dusting (C) has the highest mean yield, 13.12, and exceeds each other treatment in Tukey comparisons.

Findings and conclusion

BMI is higher on average in the prediabetes and diabetes groups, with substantial overlap between respondents.

Poor-health days have a weak positive BMI association. The multiple model explains 6.8% of BMI variation.

Outlier filtering changes estimates and health-status composition. Normal and Poisson fits are inadequate for these observations.

The experimental supplements demonstrate pairing and nuisance-factor control.

Limitations include self-report, historical data, missing survey weights, confounding, sparse subgroups and multiple exploratory tests.

The project answers descriptive and statistical questions. It supports cautious associations and course-method demonstrations.

References: CDC BRFSS 2015 | Alex Teboul (Kaggle) | MASS | agridat | NIST. Full links in notes and report.